

RESCURROUTE AI

Closed-Loop Emergency Obstruction Response Agent

Active Emergency Routing, Twilio SOS Dispatch & Automated Road Clearance Engine

Twilio Auto-SOS

Haversine Routing

Google Traffic

ElevenLabs Voice

Emergency Hackathon Presentation • Bengaluru Urban Grid

[!] Urban Gridlock & Blockages

- Ambulance delays directly correlate with increased mortality.
- Dense cities (Bengaluru) suffer constant road blockages:
 - Breakdown of heavy vehicles in narrow lanes.
 - Unannounced construction & severe traffic gridlocks.
 - Illegal parking blocking dedicated emergency vectors.
- Paramedics lose 15-25 minutes navigating bottlenecks.

[!] Passive GPS Limitations

- Standard Navigation Apps (Google Maps / Waze):
 - Only inform drivers passively after getting stuck.
 - Do NOT physically clear obstructions.
 - Route ambulances into narrow residential back-alleys.
- Lack of Direct Integration:
 - Zero real-time link between ambulances & traffic police.
 - No automated verification or responder escalation.

1. Active Road Clearance

Dispatches tow trucks & traffic police to physically clear obstructions before ambulance arrival.

2. Twilio Auto-SOS

Triggers instant HTTP REST API SMS alerts to responders upon pickup address selection.

3. Haversine Shortest AI

Calculates exact spherical distances to 12 hospitals and auto-selects the nearest facility.

4. Instant Neon Rerouting

If SLA (90s) is breached, instantly reroutes vector via clear arterial corridors (0ms delay).

Depth & Dynamic Lighting

Traffic Layer (mt1.google.com)

Basic Auth & Trial Fallback)

/ eleven_multilingual_v2)

00s SLA Countdown Timers

Presentation Layer

- HTML5 & Modern Vanilla CSS3
- Glassmorphism Design Tokens
- 3-Dashboard Responsive Layout

Maps & GIS Engine

- Leaflet.js & Mapbox GL JS v3
- Google Maps Traffic Flow Tiles
- OSRM Routing & Waypoint Engine

API Integration

- Twilio REST API (HTTP Basic Auth)
- ElevenLabs Voice AI (Rachel)
- n8n State Machine Webhook

3D Visual Engine

- Three.js WebGL Canvas
- Interactive Mouse Parallax Depth
- Dynamic Lighting & Materials

SMS.

cross 12 hospitals.

ection.

